

Z → ττ IN FOUR CHANNELS

CMS Open Data 2016 (Run2016G+H), 16.4 fb⁻¹ — τ_hτ_h + μτ_h + eτ_h + eμ, fitted together 17 September 2026

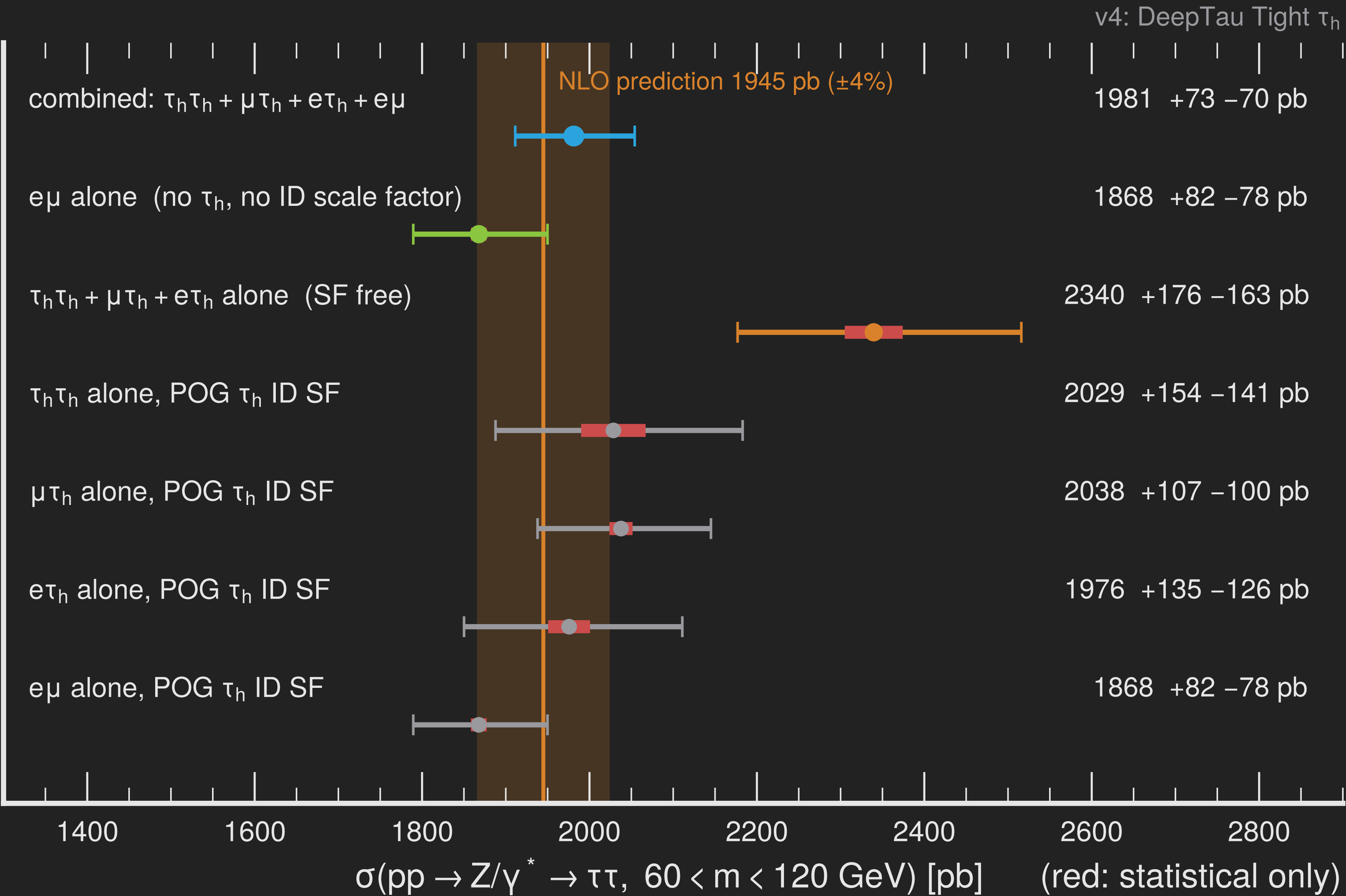
sigma(pp -> Z/gamma* -> tautau, 60 < m < 120 GeV) = 1981 +73 -70 pb (NLO prediction 1945 pb)

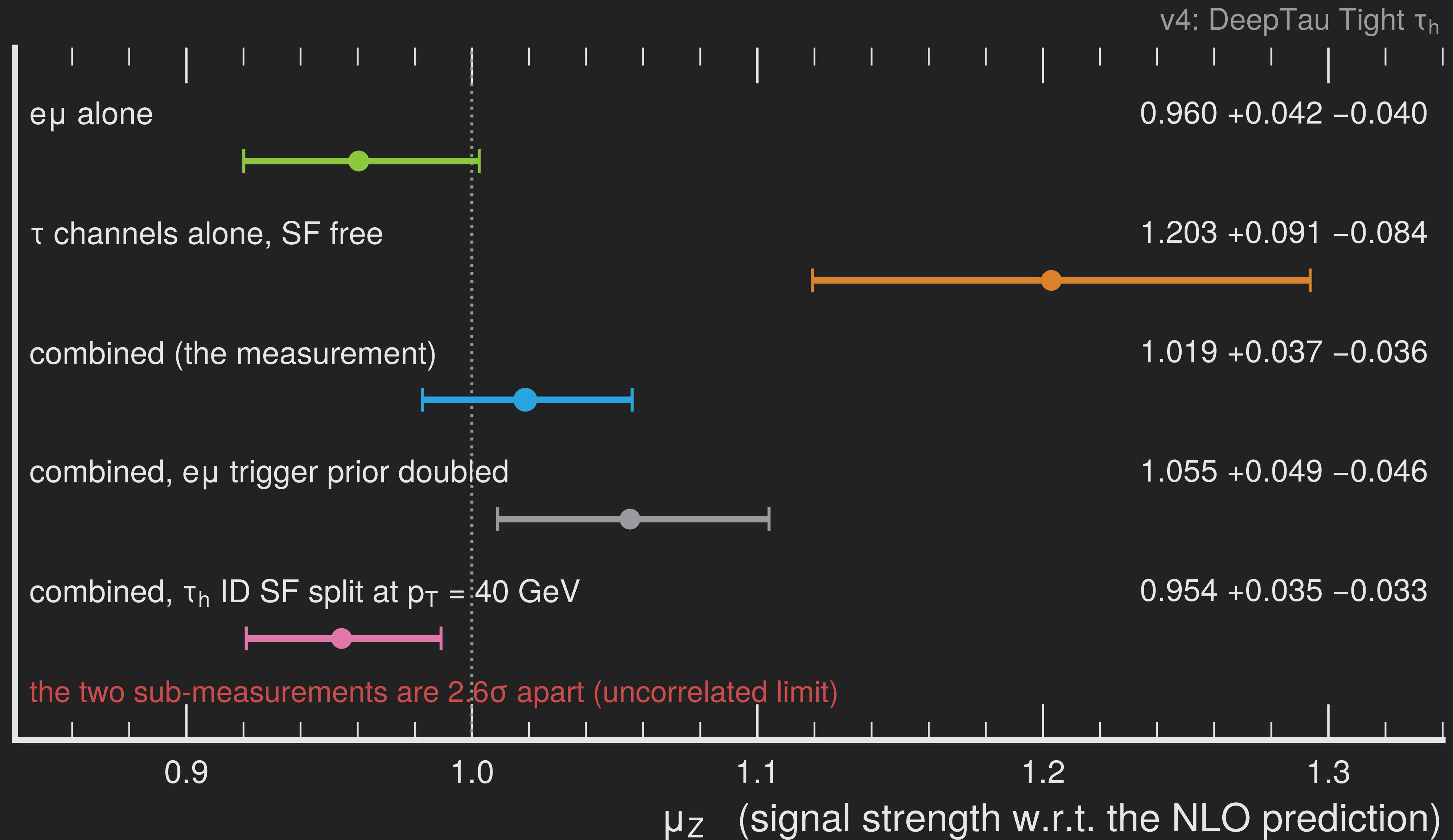
mu_Z = 1.019 +0.037 -0.036, expected +-0.036, goodness of fit p = 0.15, mu_ttbar = 1.11

The tau_h identification scale factors and the tau_h energy scale are measured in situ by this fit, not taken from the TauPOG. Jet -> tau_h fakes come from data in every channel. There is no mu mu channel and every channel vetoes a second lepton, so this is orthogonal to Z -> mu mu and Z -> ee.

It is the compromise between two sub-measurements 2.6 sigma apart: e mu alone (no tau_h) 0.960 +0.042 -0.040, the three tau channels alone, scale factors free, 1.203 +0.091 -0.084. Both are quoted throughout.

1. THE RESULT





Every fit of the measurement

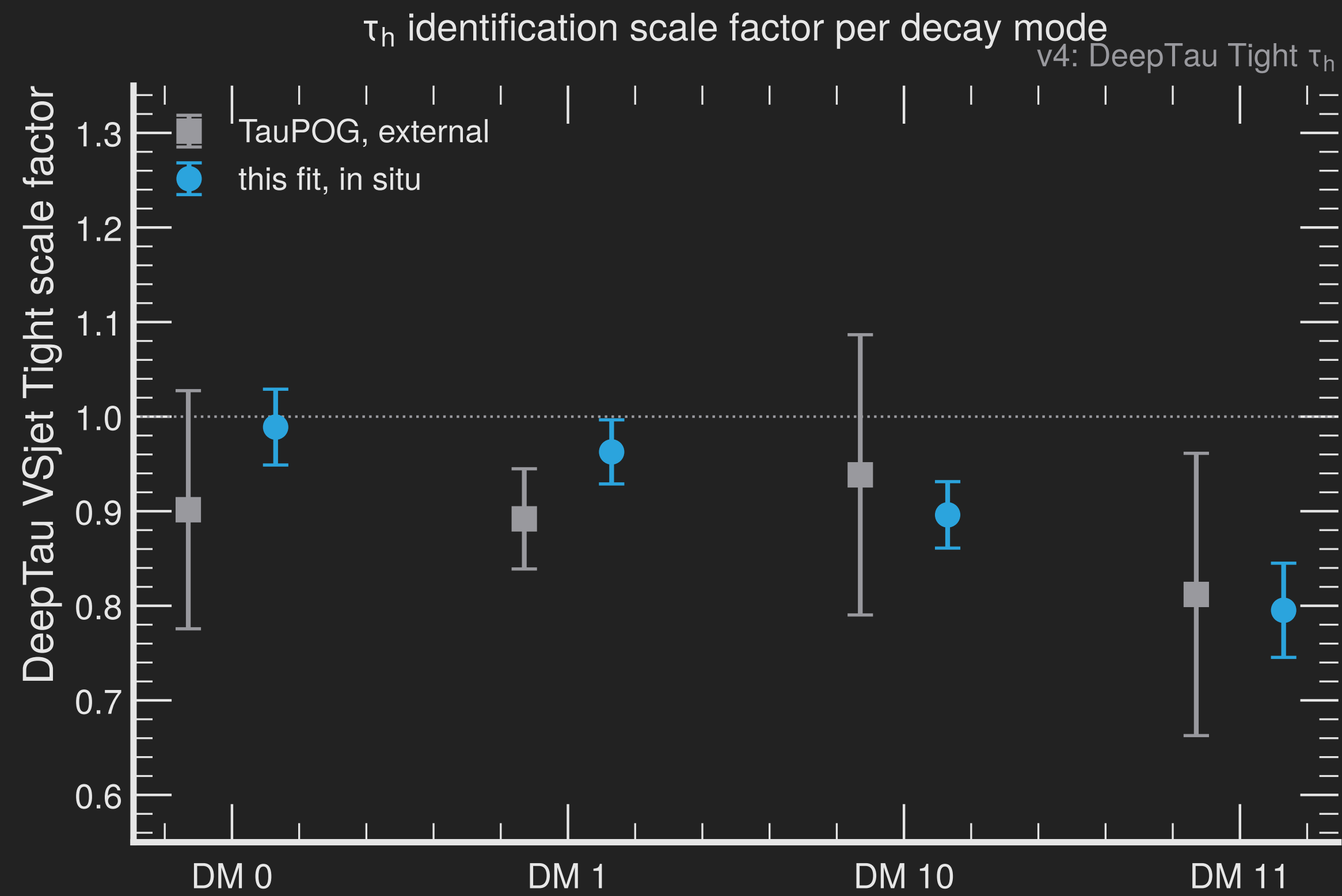
fit	μ_z	$\sigma(60-120)$ [pb]	GoF p
four channels (the measurement)	1.019 +0.037 -0.036	1981 +73 -70	0.15
$e\mu$ alone (no τ_h)	0.960 +0.042 -0.040	1868 +82 -78	0.03
τ channels alone, SF free	1.203 +0.091 -0.084	2340 +176 -163	0.48
$\tau_h\tau_h$ alone, POG SF	1.043 +0.079 -0.072	2029 +154 -141	0.33
$\mu\tau_h$ alone, POG SF	1.048 +0.055 -0.051	2038 +107 -100	0.16
$e\tau_h$ alone, POG SF	1.016 +0.070 -0.065	1976 +135 -126	0.49
$e\mu$ alone, POG SF	0.960 +0.042 -0.040	1868 +82 -78	0.03

Cross-check fits: how much the modelling choices are worth

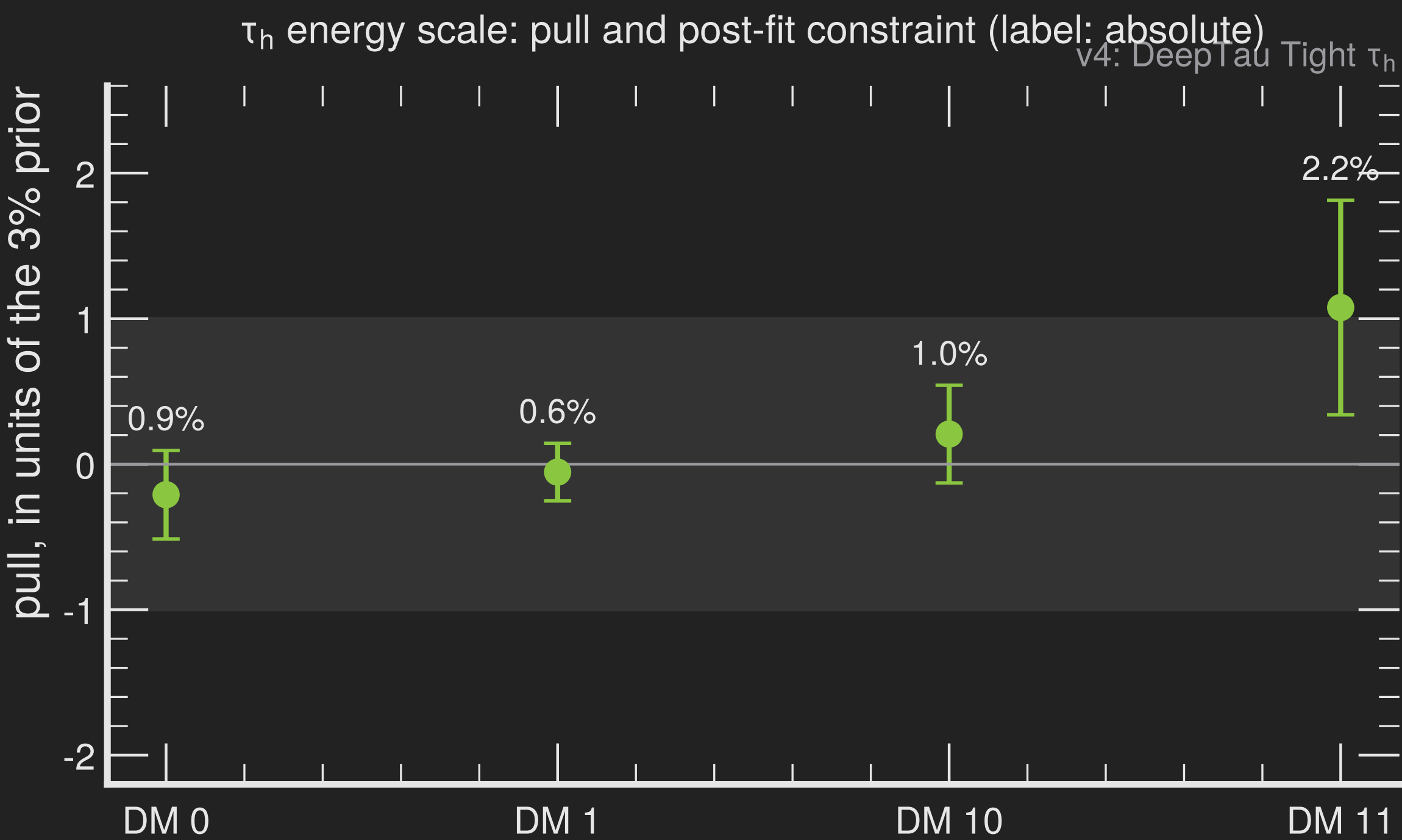
fit	μ_z	$\sigma(60-120)$ [pb]
$e\mu$ trigger prior doubled	1.055 +0.049 -0.046	2053 +95 -90
τ_h ID SF split at 40 GeV	0.954 +0.035 -0.033	1856 +68 -65

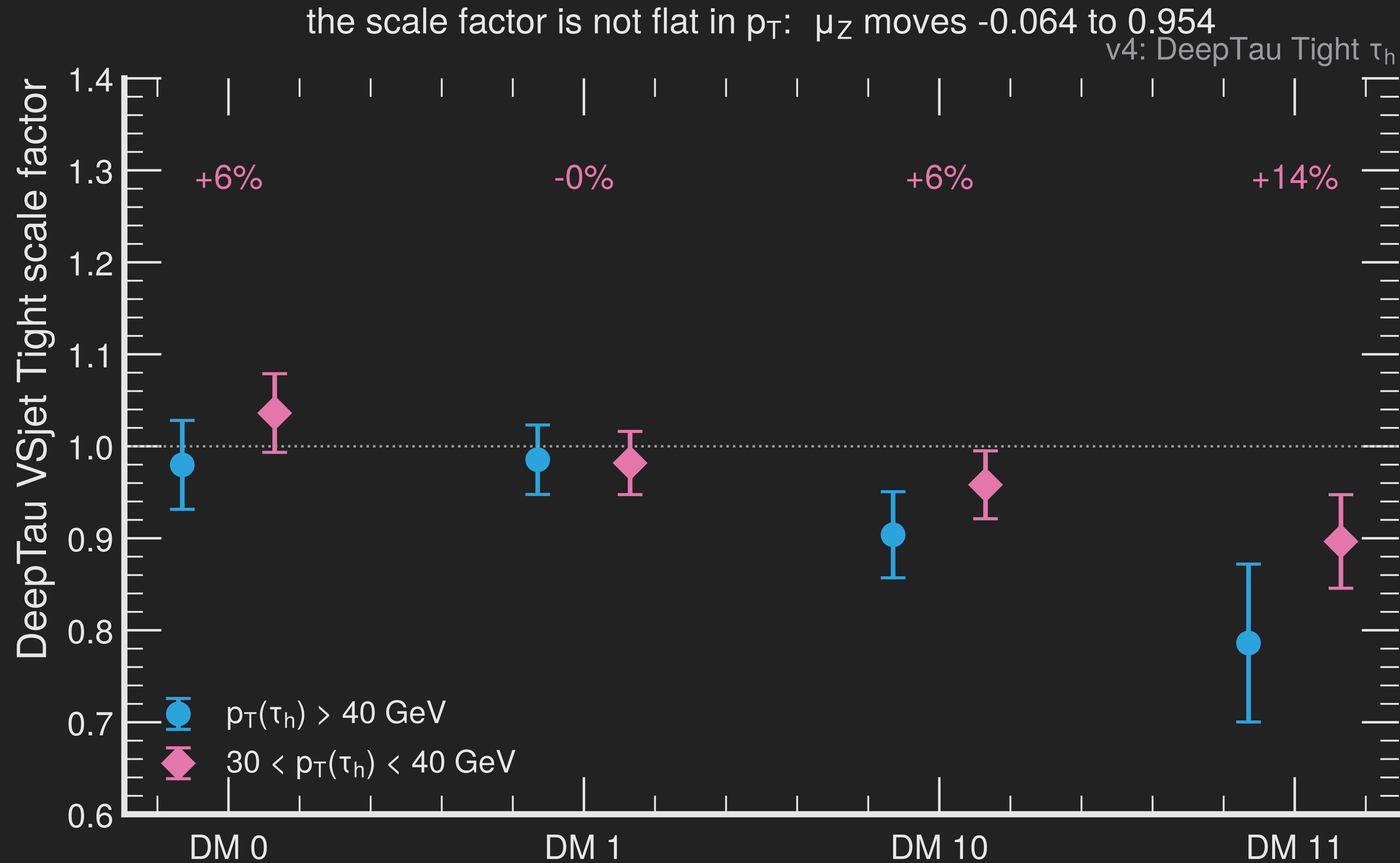
2. WHAT THE FIT MEASURES IN SITU

fitted against the TauPOG values, which are only priors here



the 3% prior is constrained to 0.6–2.2% by the $m_{\tau\tau}$ shapes





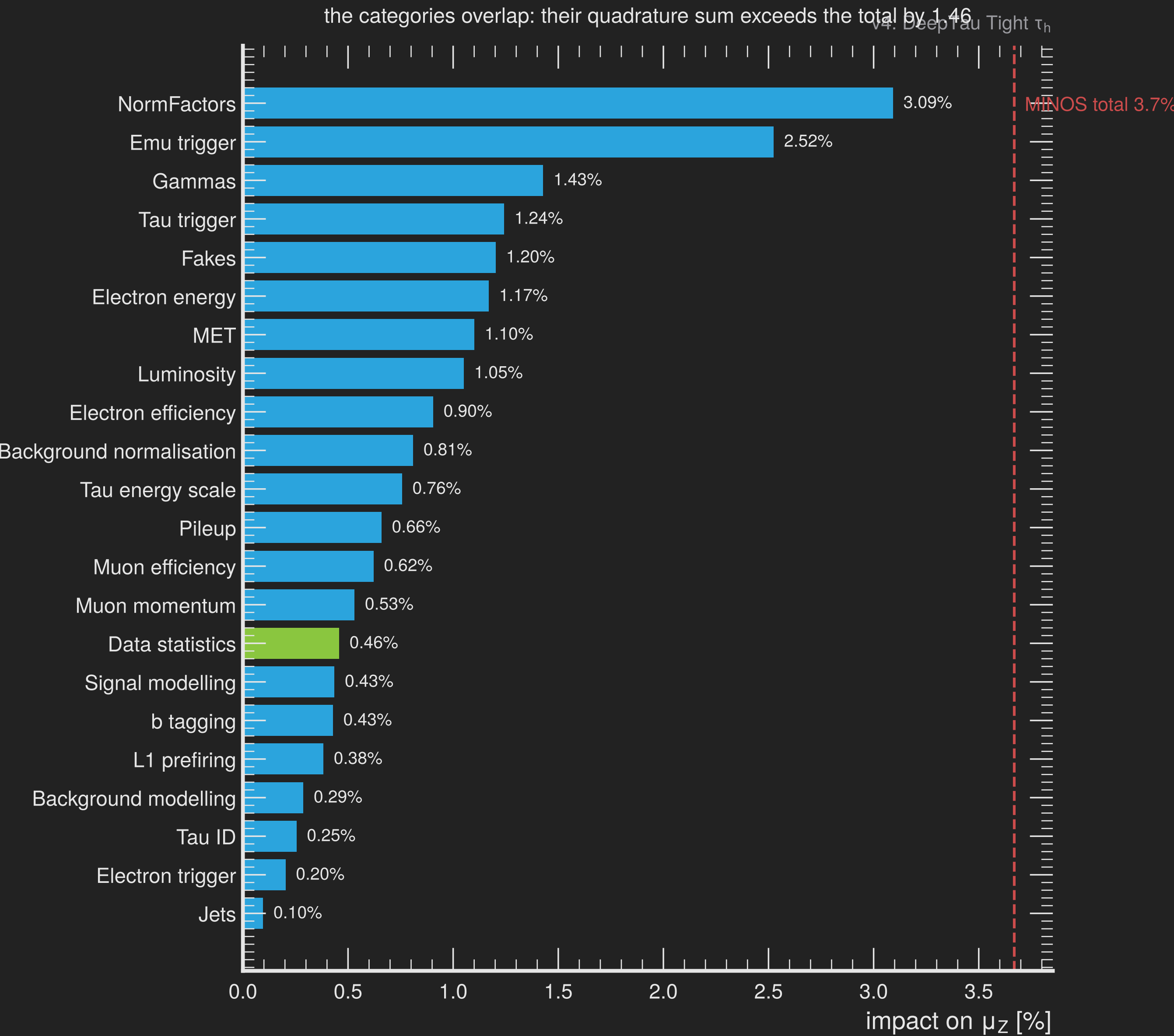
τ_h identification scale factor (DeepTau VSjet Tight)

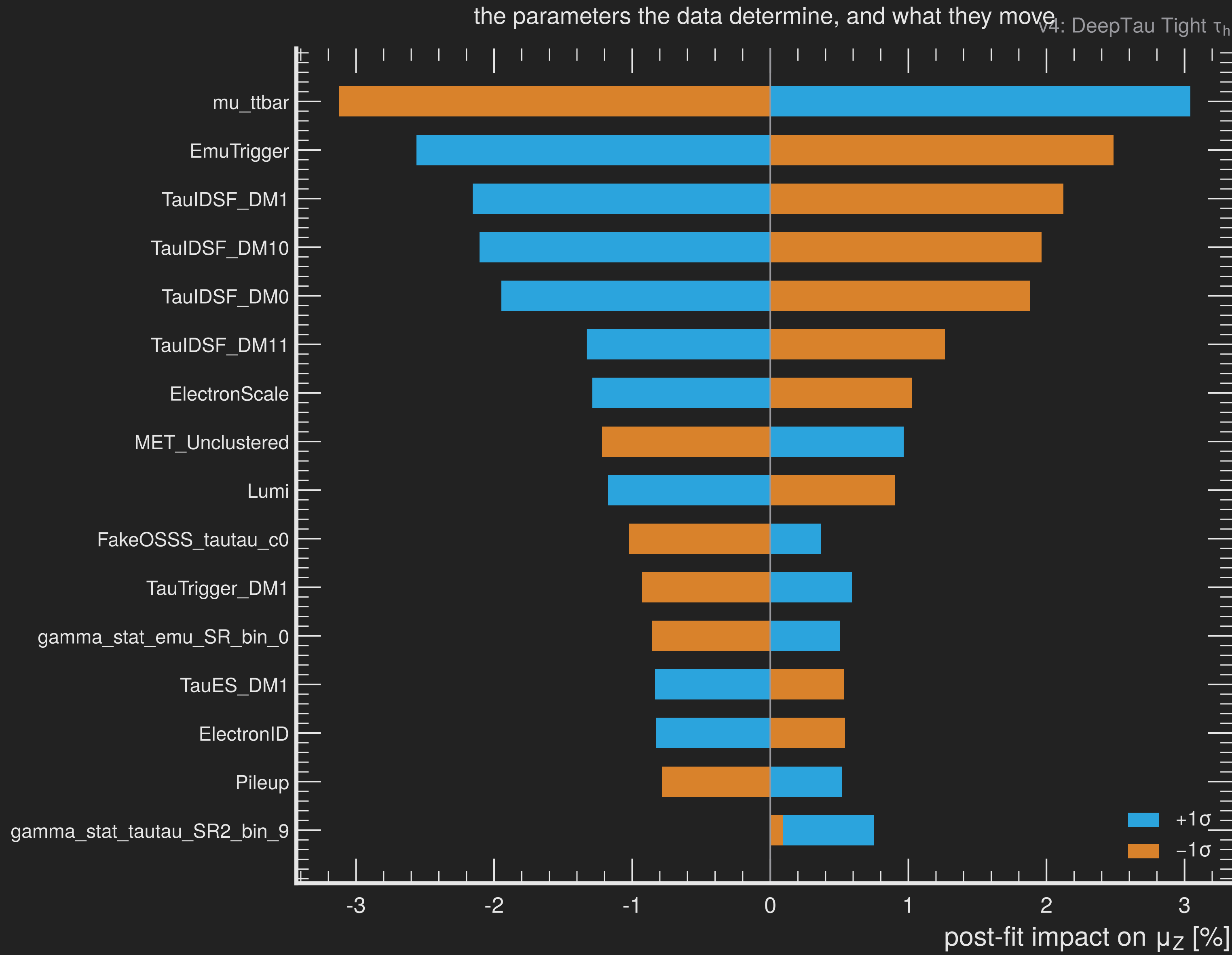
	this fit	TauPOG (prior only)	precision
decay mode 0	0.989 ± 0.040	0.902 ± 0.126	4.1%
decay mode 1	0.963 ± 0.034	0.892 ± 0.053	3.5%
decay mode 10	0.896 ± 0.035	0.938 ± 0.148	3.9%
decay mode 11	0.795 ± 0.050	0.812 ± 0.149	6.3%

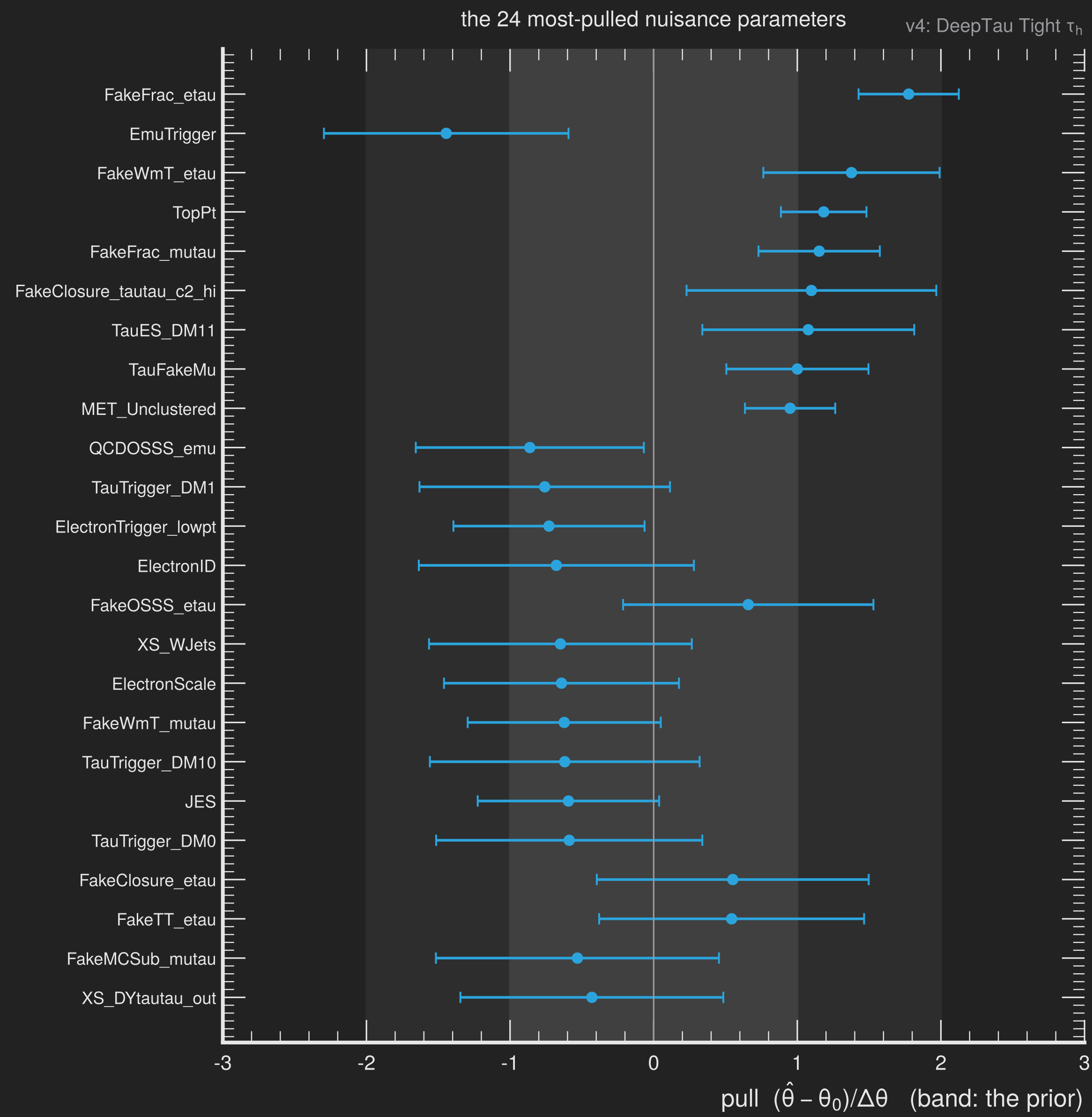
τ_h energy scale, relative to the POG central value (prior $\pm 3\%$)

	shift	uncertainty	fraction of the prior
decay mode 0	-0.63%	$\pm 0.91\%$	0.30
decay mode 1	-0.16%	$\pm 0.59\%$	0.20
decay mode 10	+0.62%	$\pm 1.01\%$	0.34
decay mode 11	+3.23%	$\pm 2.21\%$	0.74

3. UNCERTAINTIES





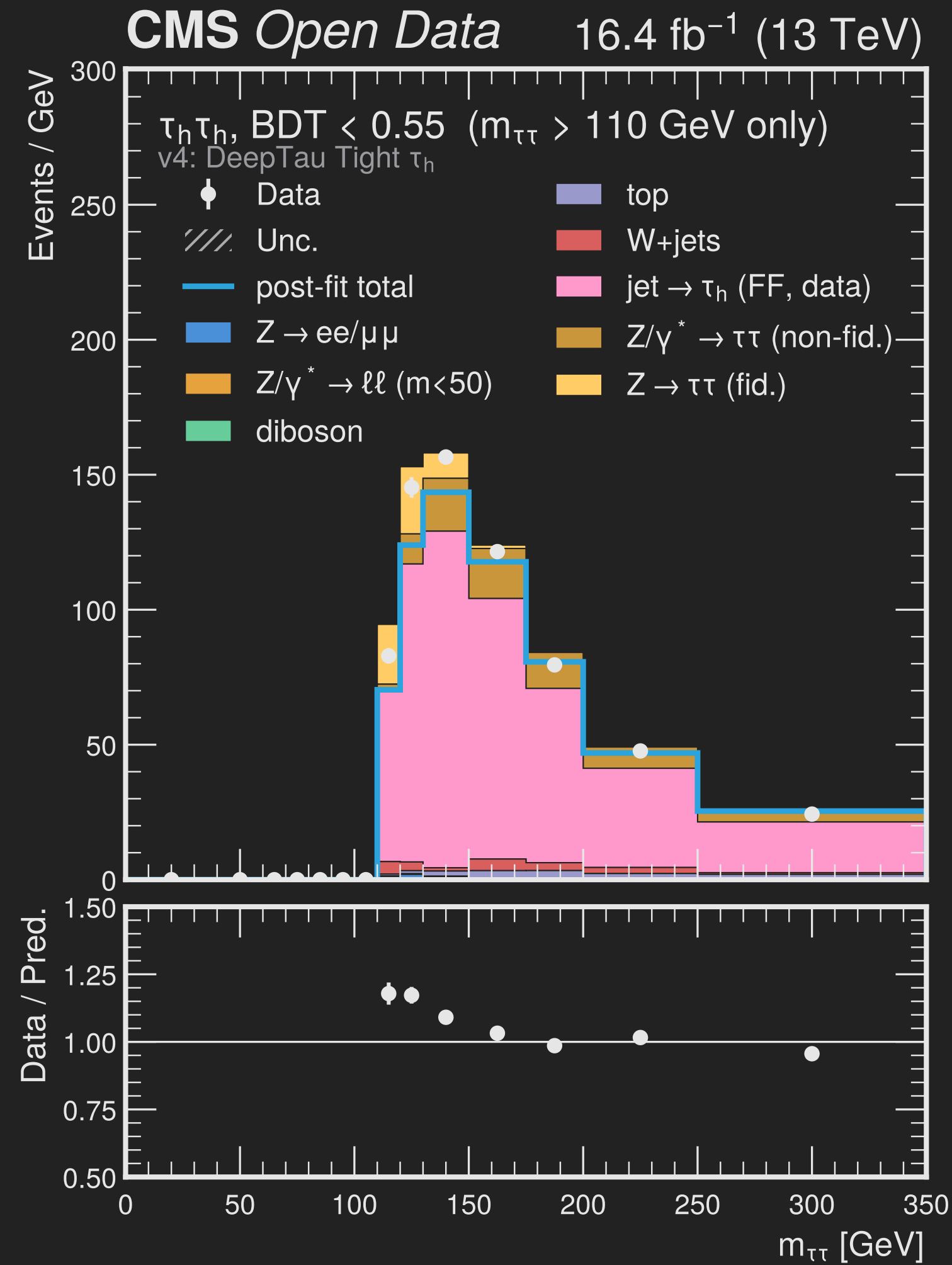


grouped impacts: their quadrature sum exceeds the MINOS total (3.7%) by 1.46, so rescale by 0.685 if a breakdown is needed

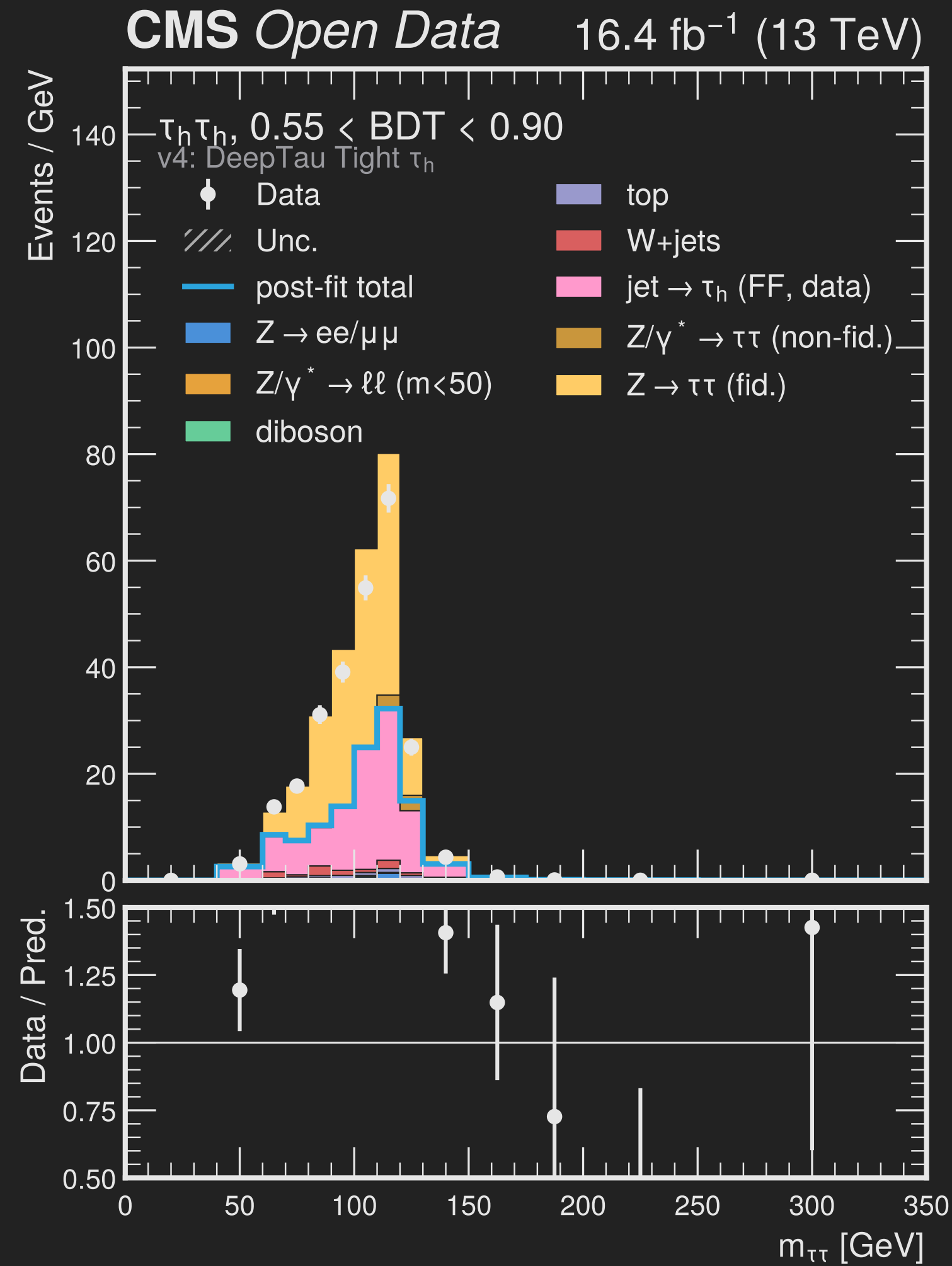
category	impact on μ_z	rescaled	ρ with $\mu\mu$ / ee
NormFactors	3.09%	2.12%	0
Emu trigger	2.52%	1.73%	0
Gammas	1.43%	0.98%	0
Tau trigger	1.24%	0.85%	0
Fakes	1.20%	0.82%	0
Electron energy	1.17%	0.80%	0
MET	1.10%	0.75%	0
Luminosity	1.05%	0.72%	1
Electron efficiency	0.90%	0.62%	0
Background normalisation	0.81%	0.55%	1
Tau energy scale	0.76%	0.52%	0
Pileup	0.66%	0.45%	1
Muon efficiency	0.62%	0.42%	0
Muon momentum	0.53%	0.36%	0
Signal modelling	0.43%	0.30%	1
b tagging	0.43%	0.29%	0
L1 prefiring	0.38%	0.26%	1
Background modelling	0.29%	0.20%	0
Tau ID	0.25%	0.17%	0
Electron trigger	0.20%	0.14%	0
Jets	0.10%	0.07%	0
data statistics	0.46%	0.31%	0

4. THE 13 FITTED REGIONS

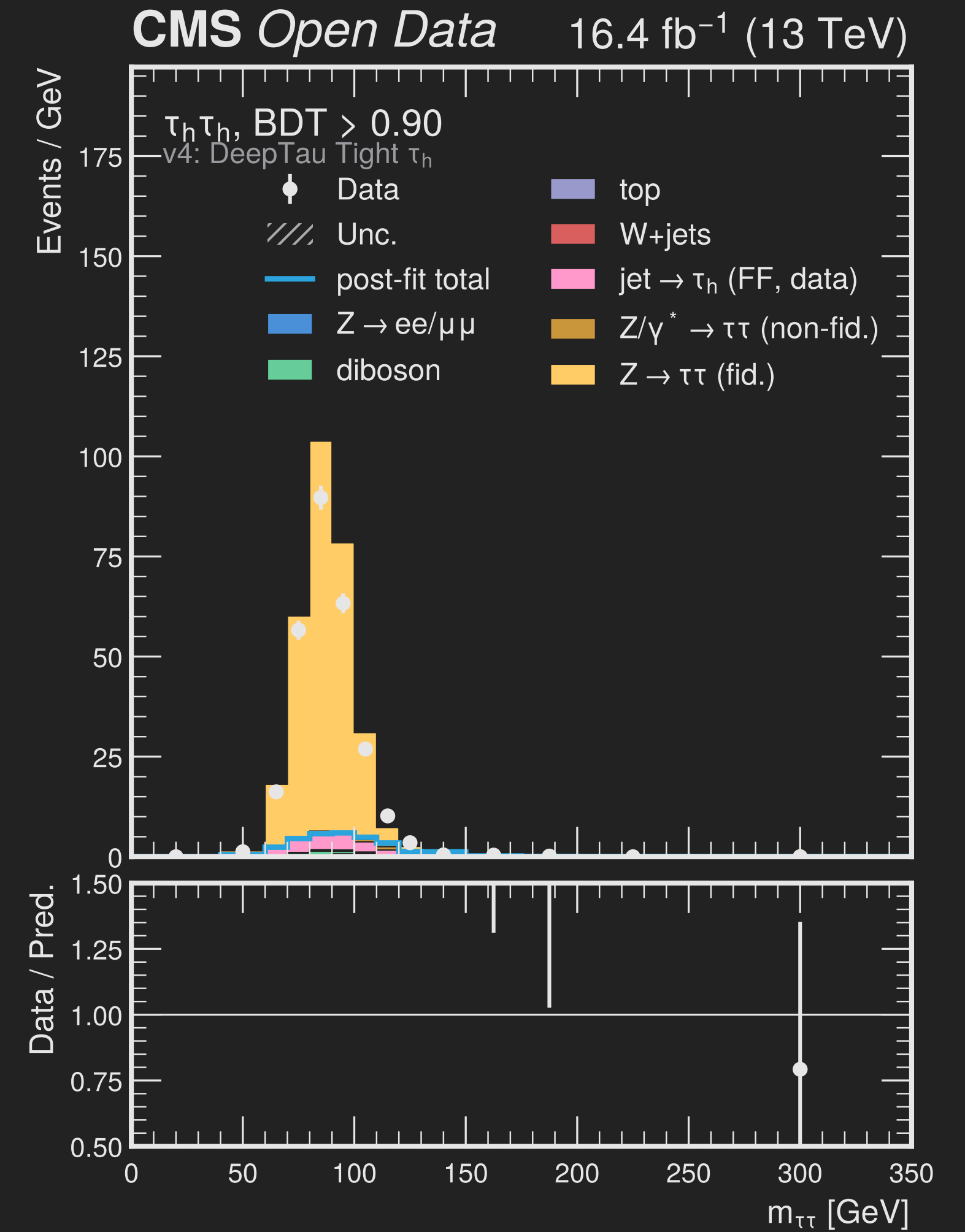
$\tau_h\tau_h$, BDT < 0.55 (fake sideband, fitted above 110 GeV)



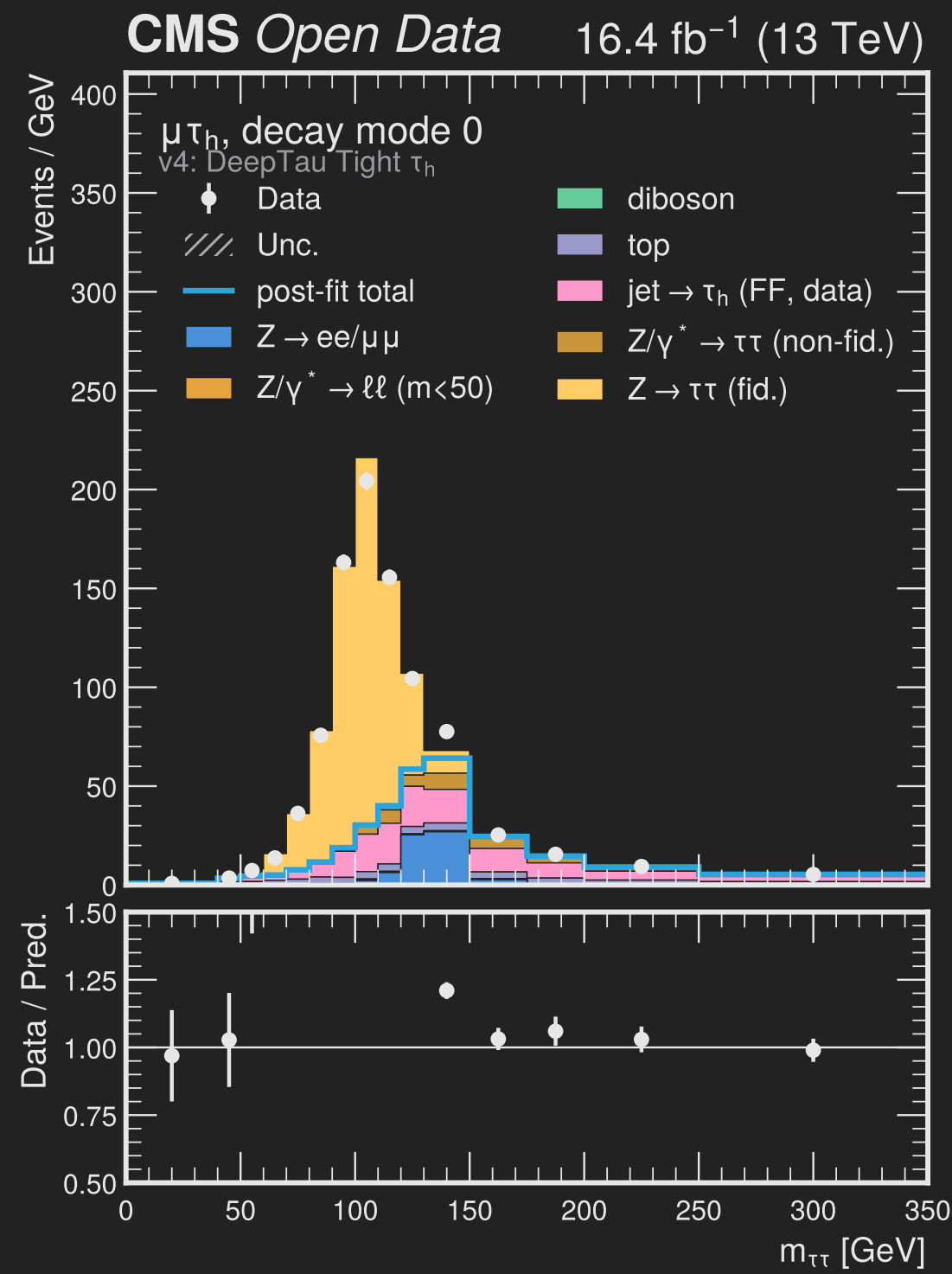
$\tau_h\tau_h$, 0.55 < BDT < 0.90



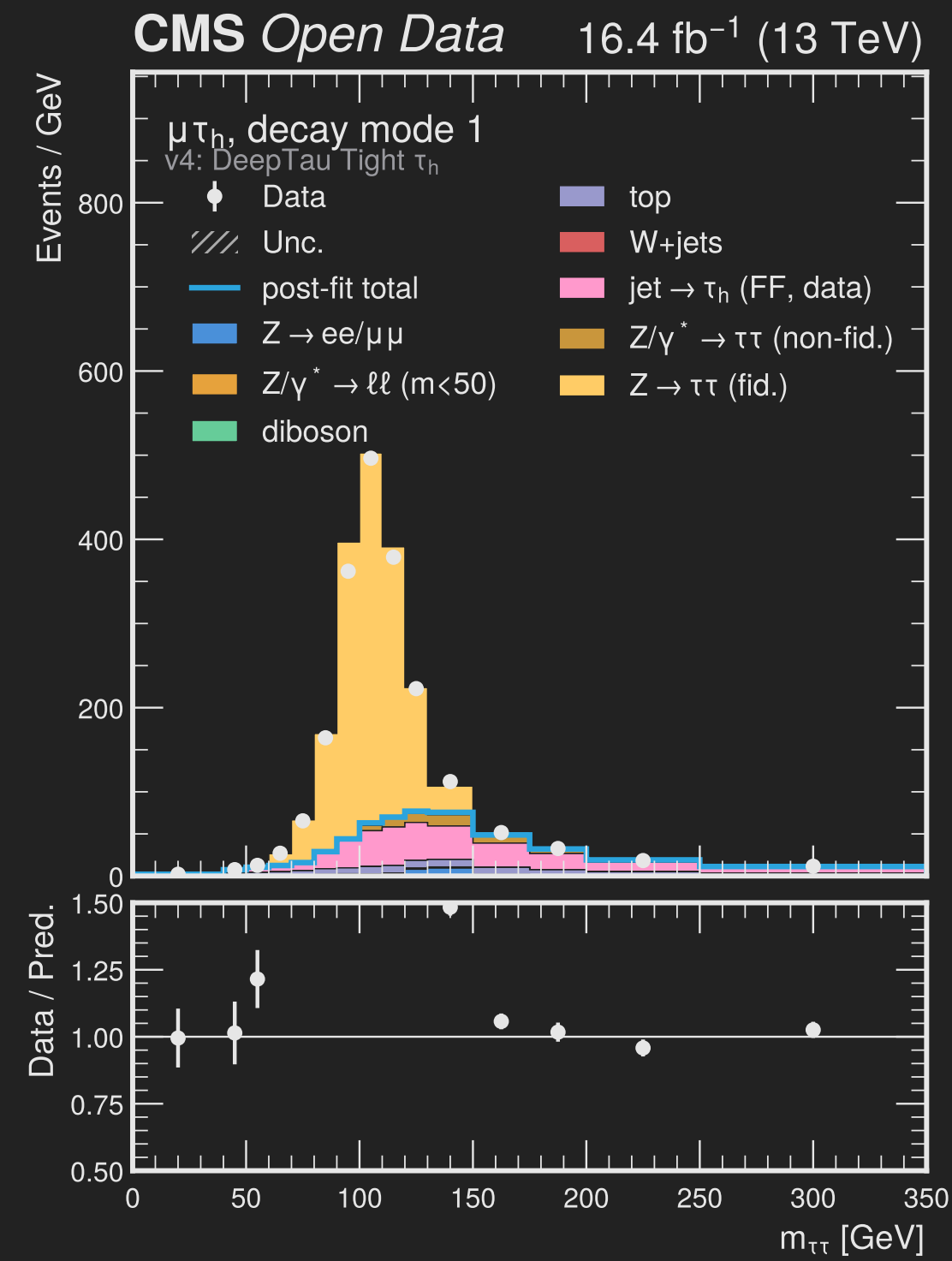
$\tau_h\tau_h$, BDT > 0.90



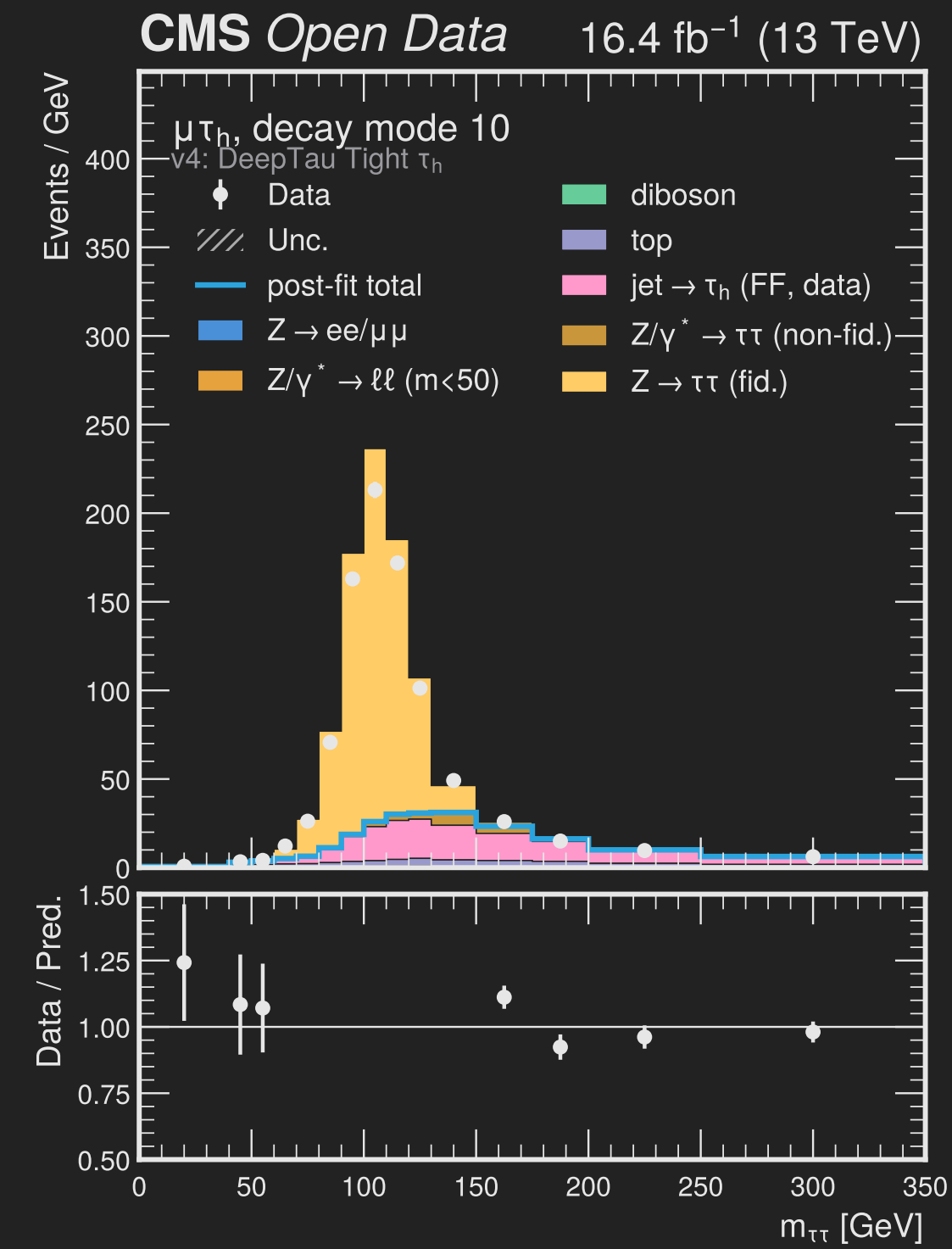
decay mode 0



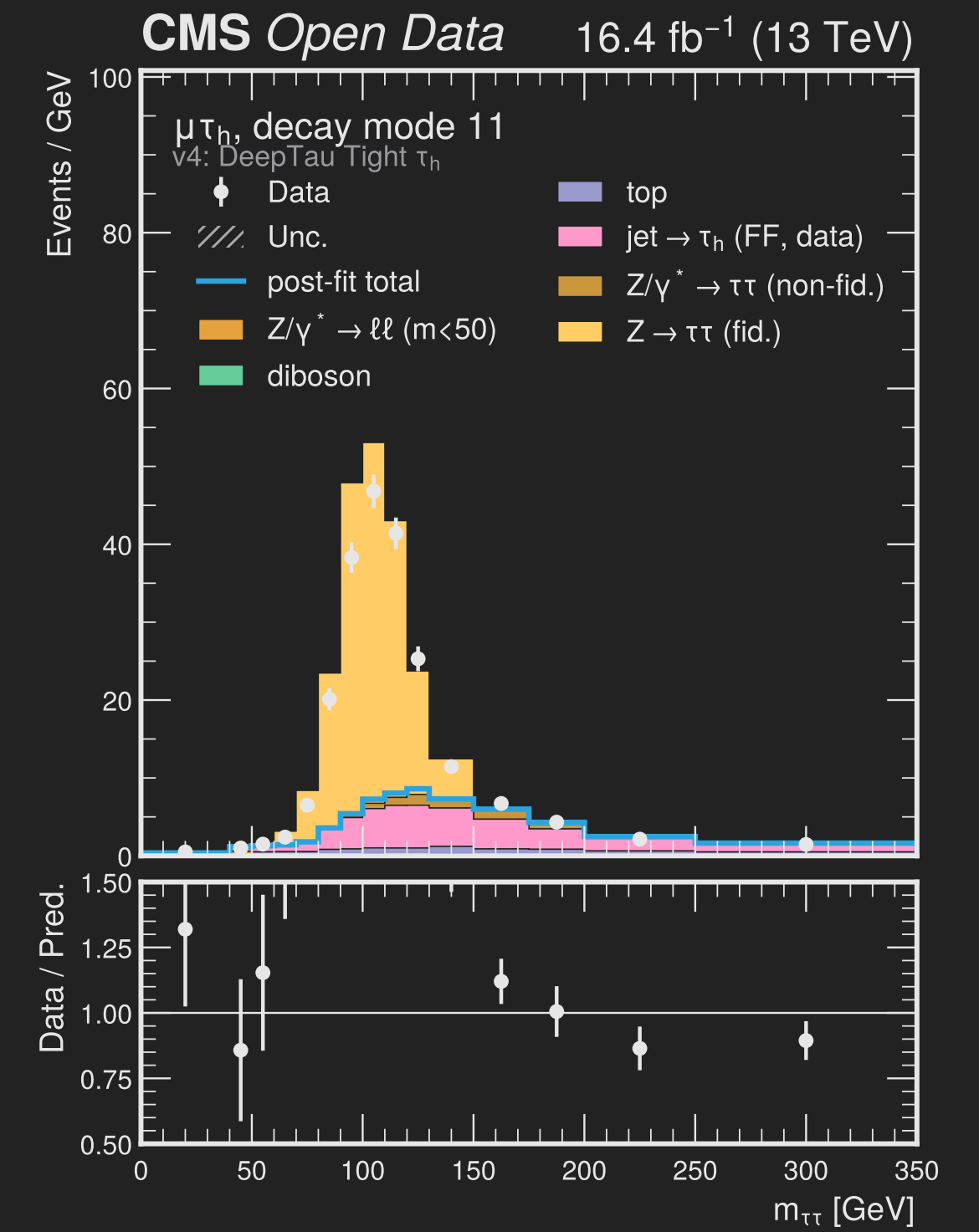
decay mode 1



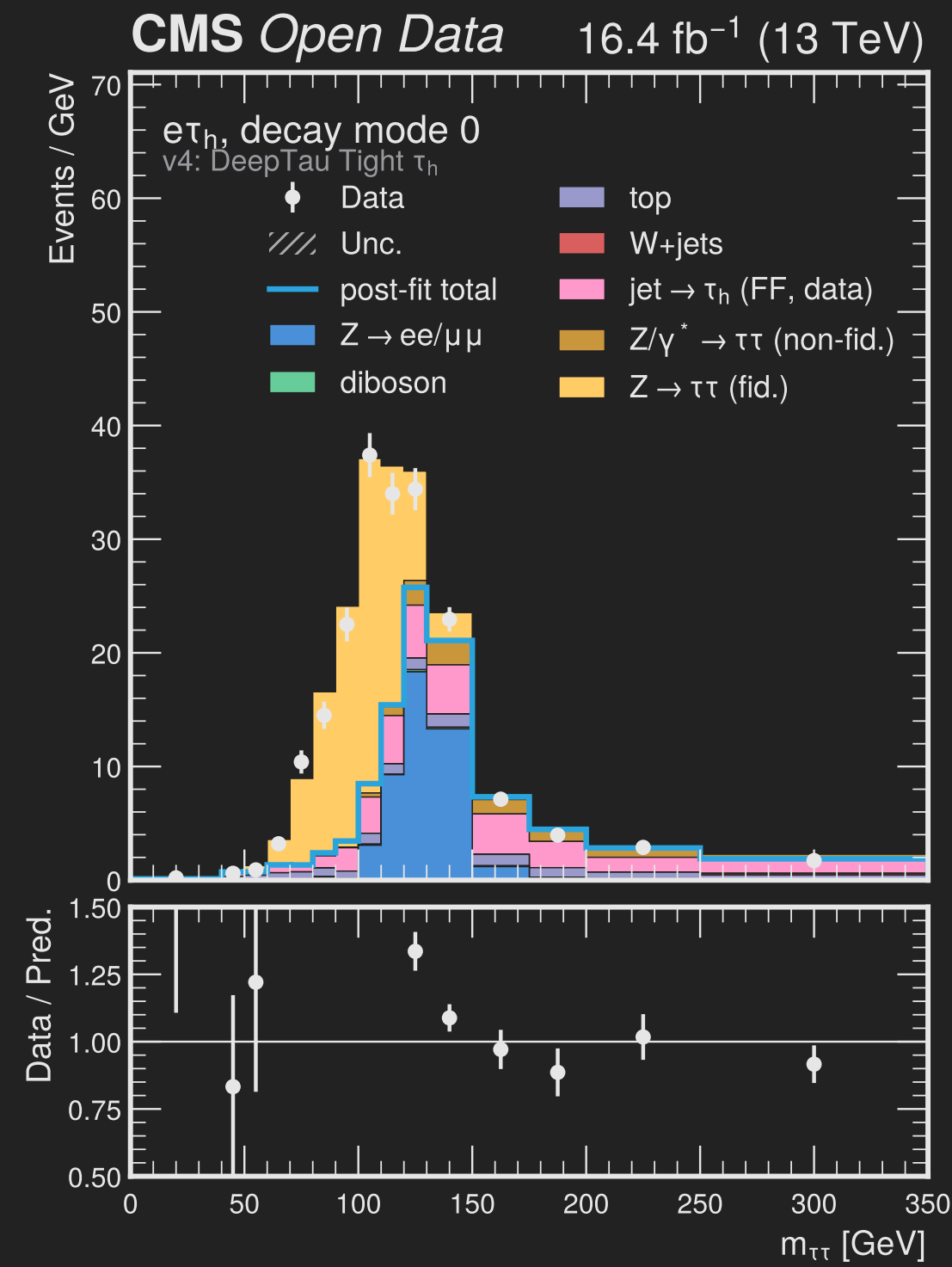
decay mode 10



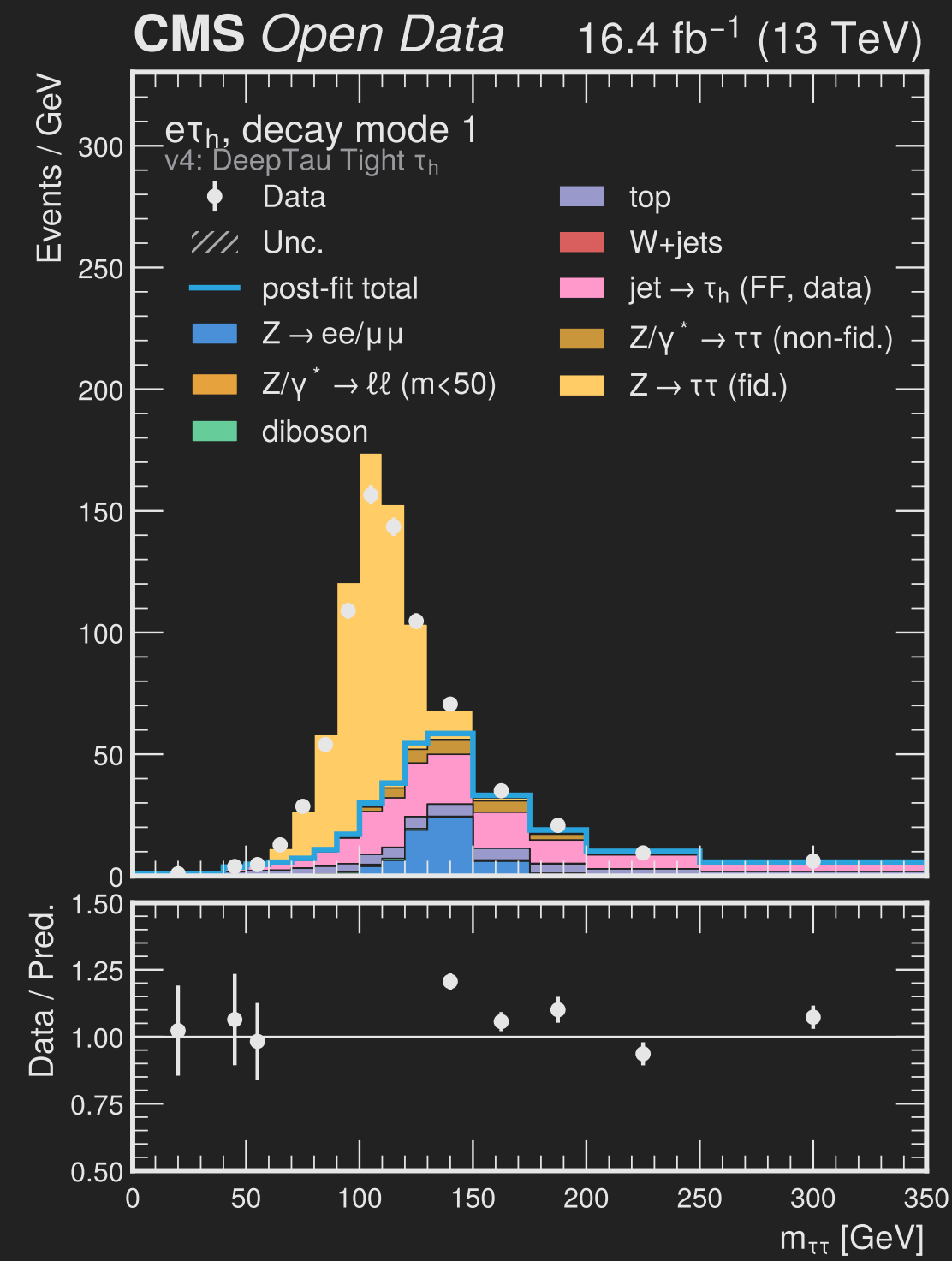
decay mode 11



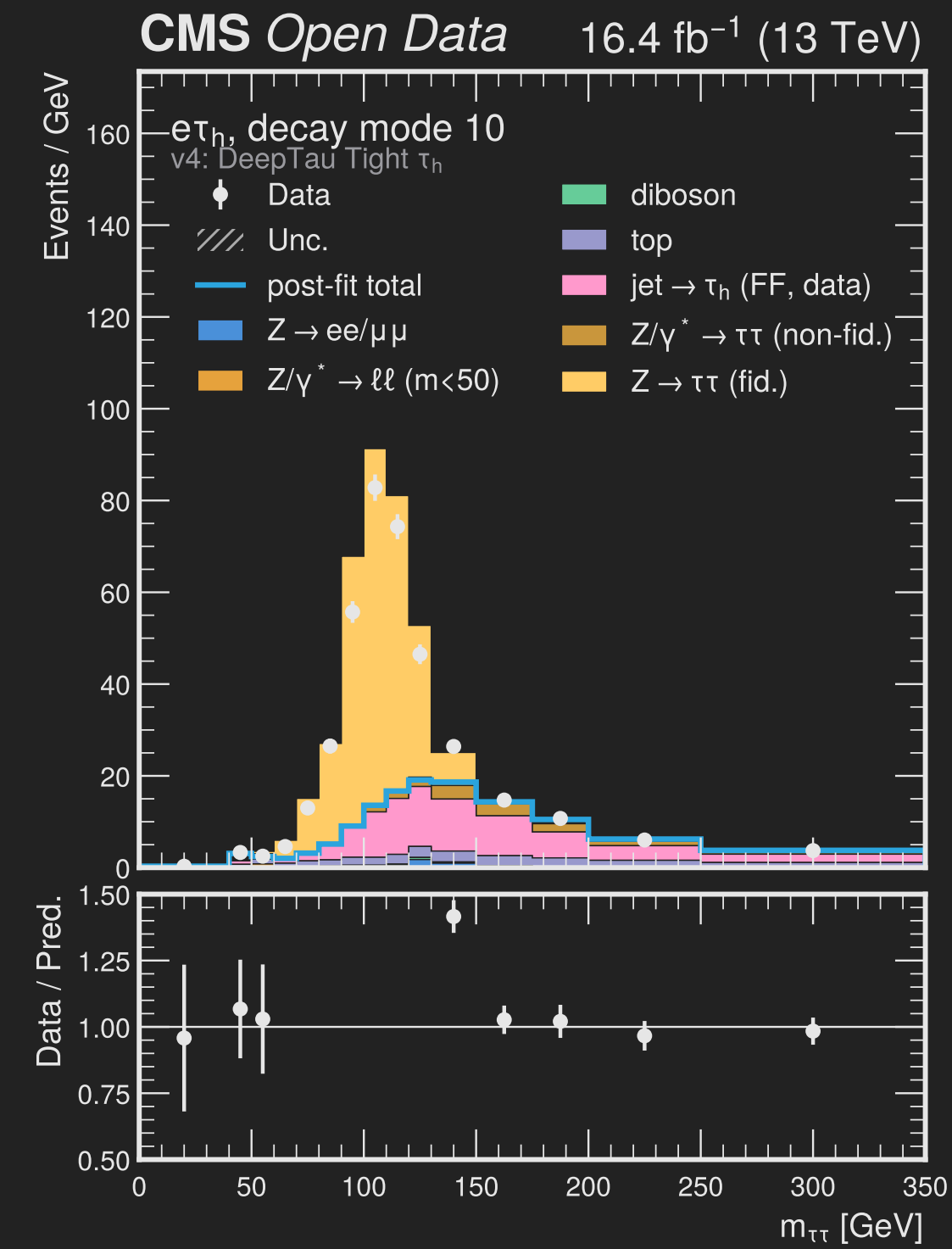
decay mode 0



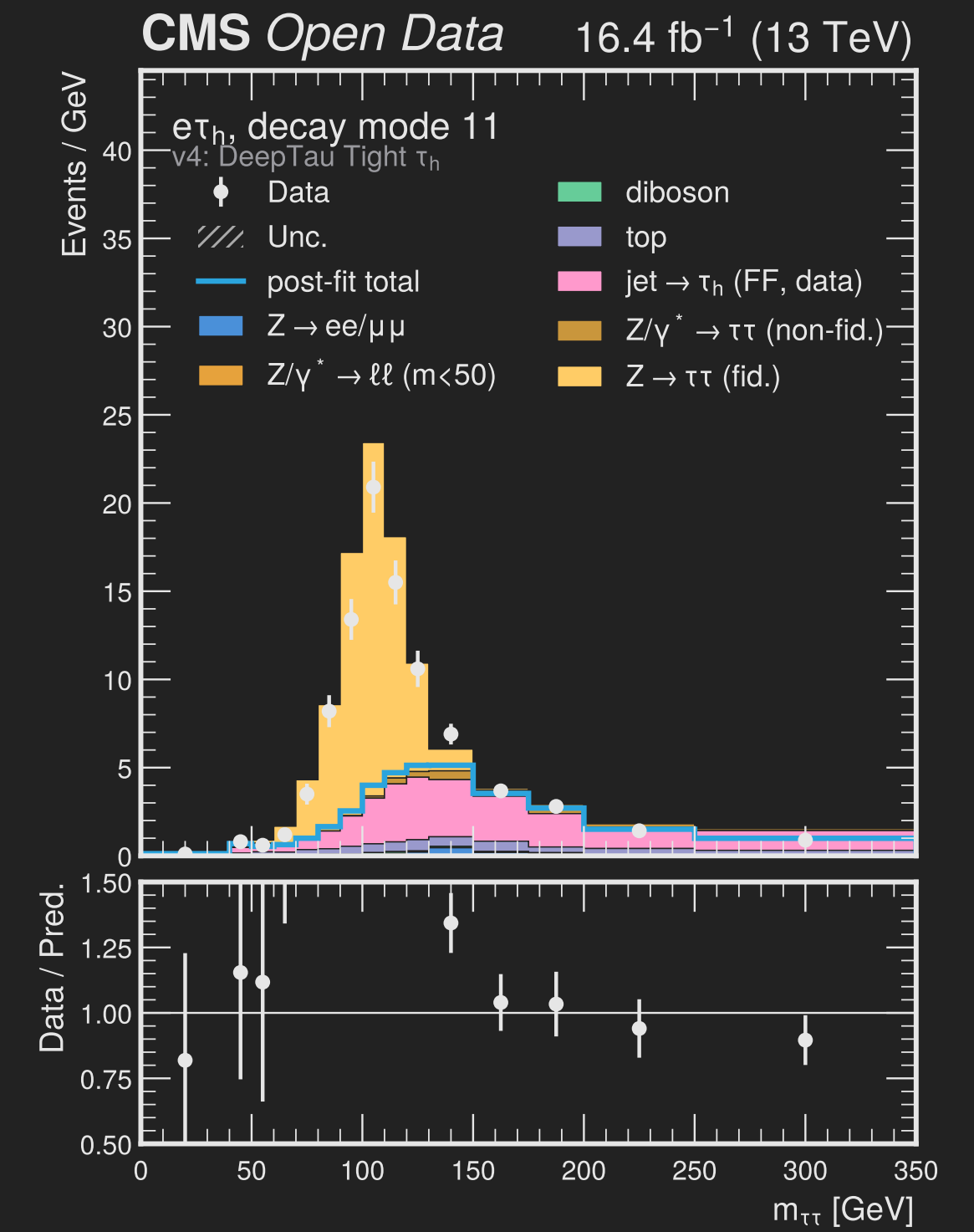
decay mode 1

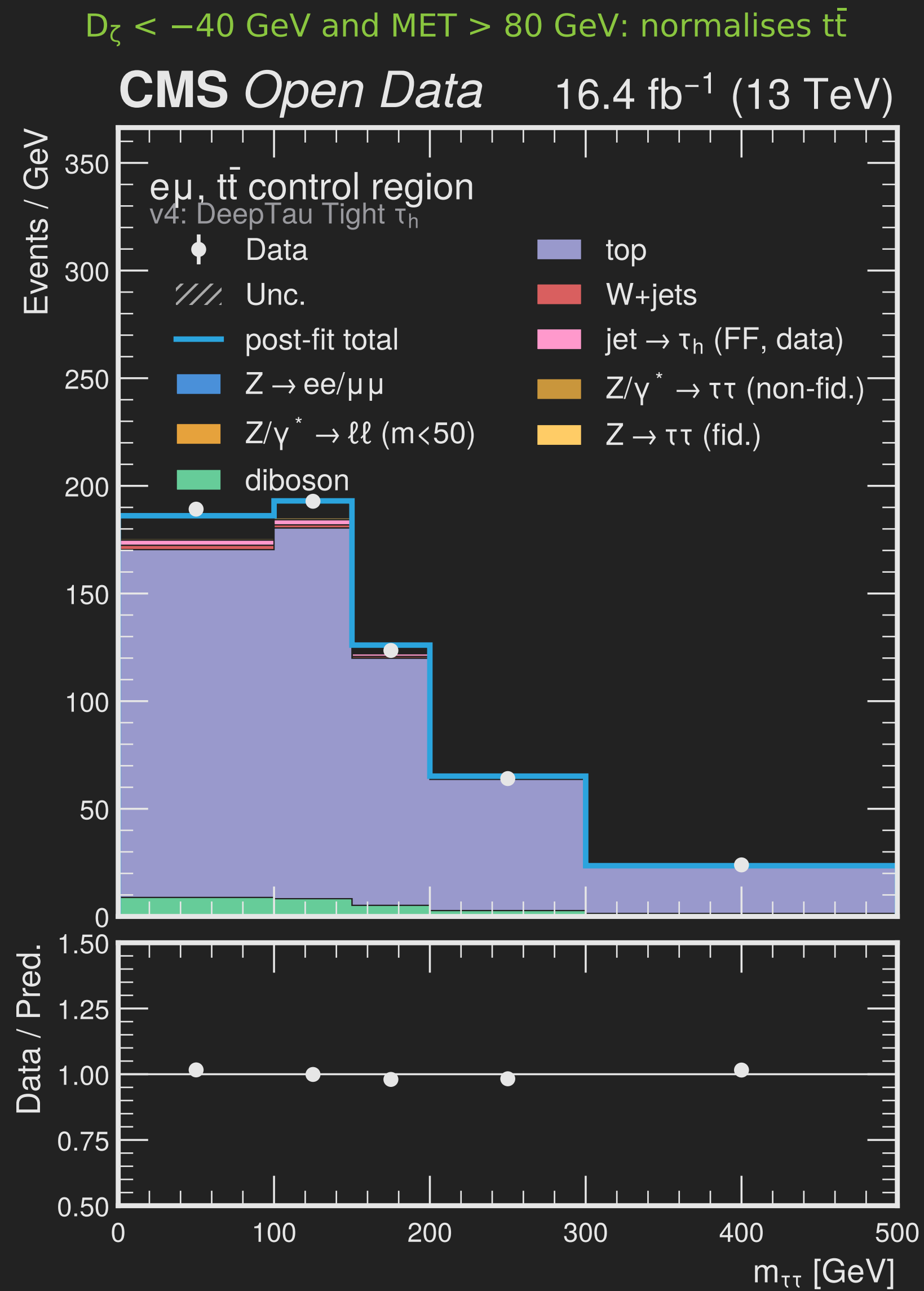
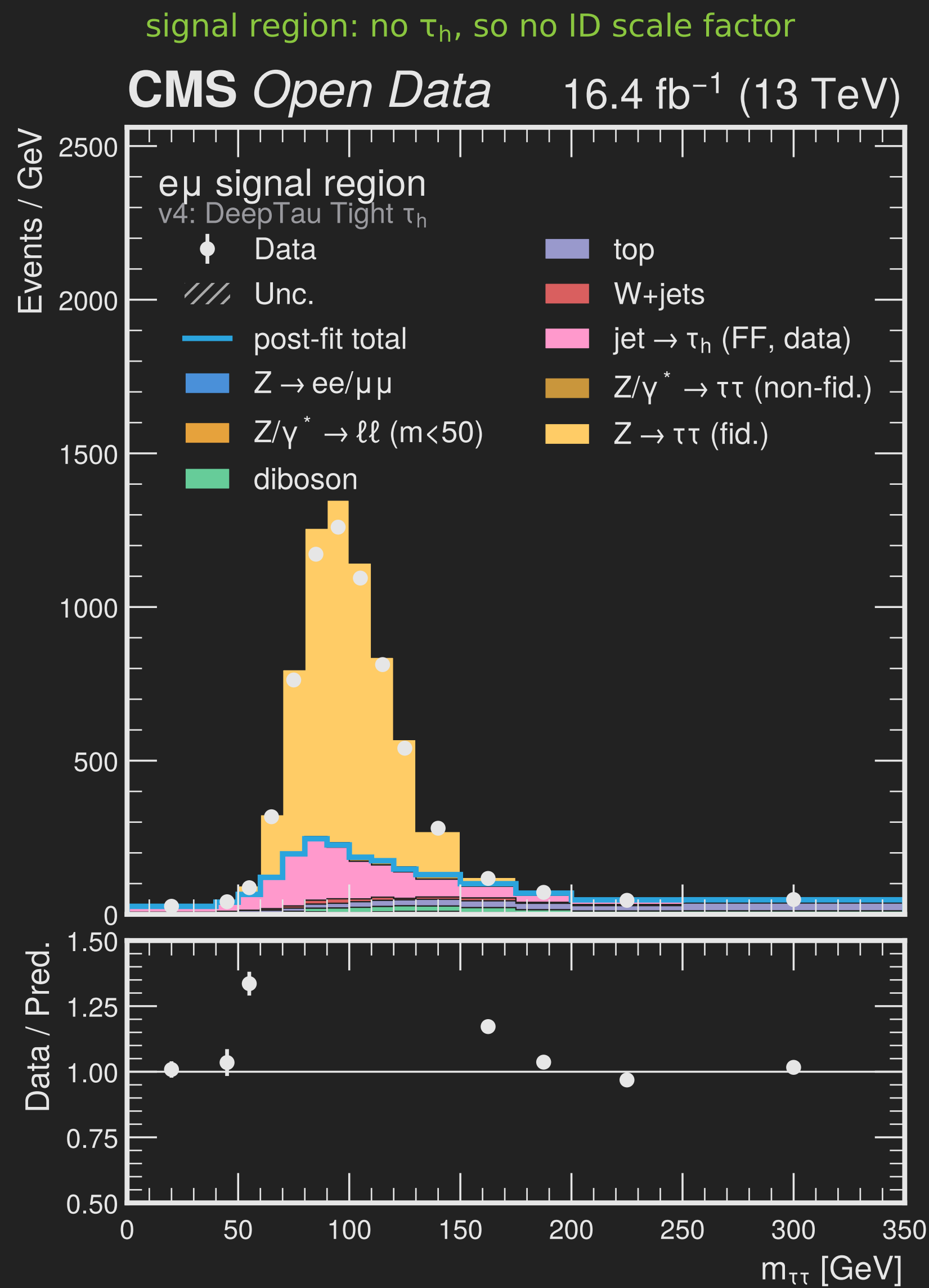


decay mode 10



decay mode 11



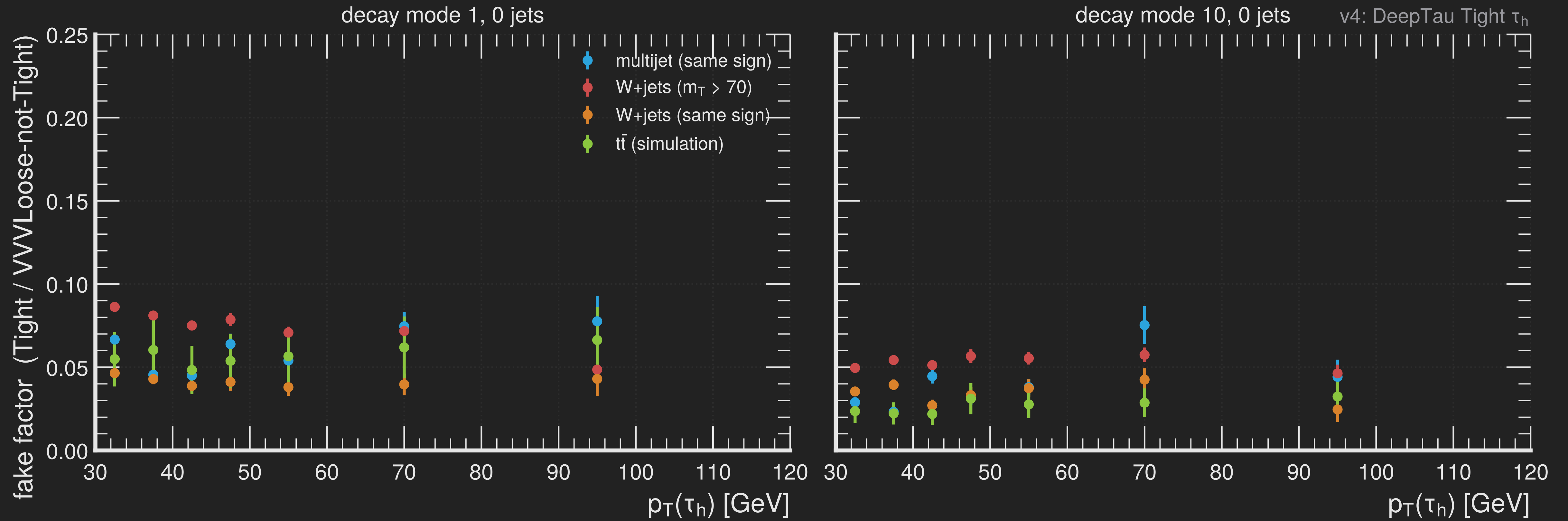


events, before the fit

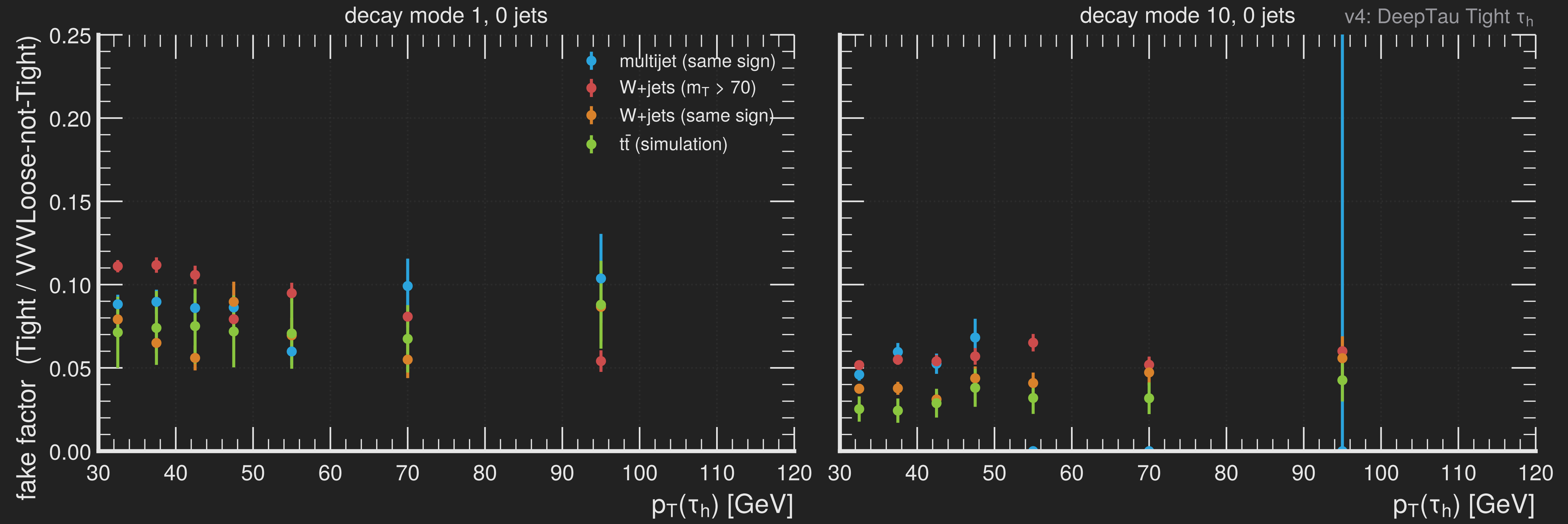
region	Data	DYtautau	Fakes	TTbar	DYtautau_out	DYee	DYmumu	WJets
tautau_SR0	15742	795	12351	425	2239	68	0	394
tautau_SR1	2704	1630	1044	48	73	26	0	69
tautau_SR2	2714	2749	197	19	50	16	0	-88
emu_SR	79300	52985	15241	5342	2040	189	499	1401
emu_CRtt	45931	116	406	37650	9	1	16	352
mutau_SR_dm0	11244	6313	2231	673	735	0	975	0
mutau_SR_dm1	23866	15545	5309	1550	1295	1	324	8
mutau_SR_dm10	10795	7316	2797	689	584	0	7	0
mutau_SR_dm11	2611	1794	710	175	151	0	0	0
etau_SR_dm0	2639	1113	586	185	224	614	0	15
etau_SR_dm1	10106	5206	2504	820	571	958	0	10
etau_SR_dm10	4940	2877	1429	422	330	47	0	0
etau_SR_dm11	1211	692	492	104	60	20	0	0

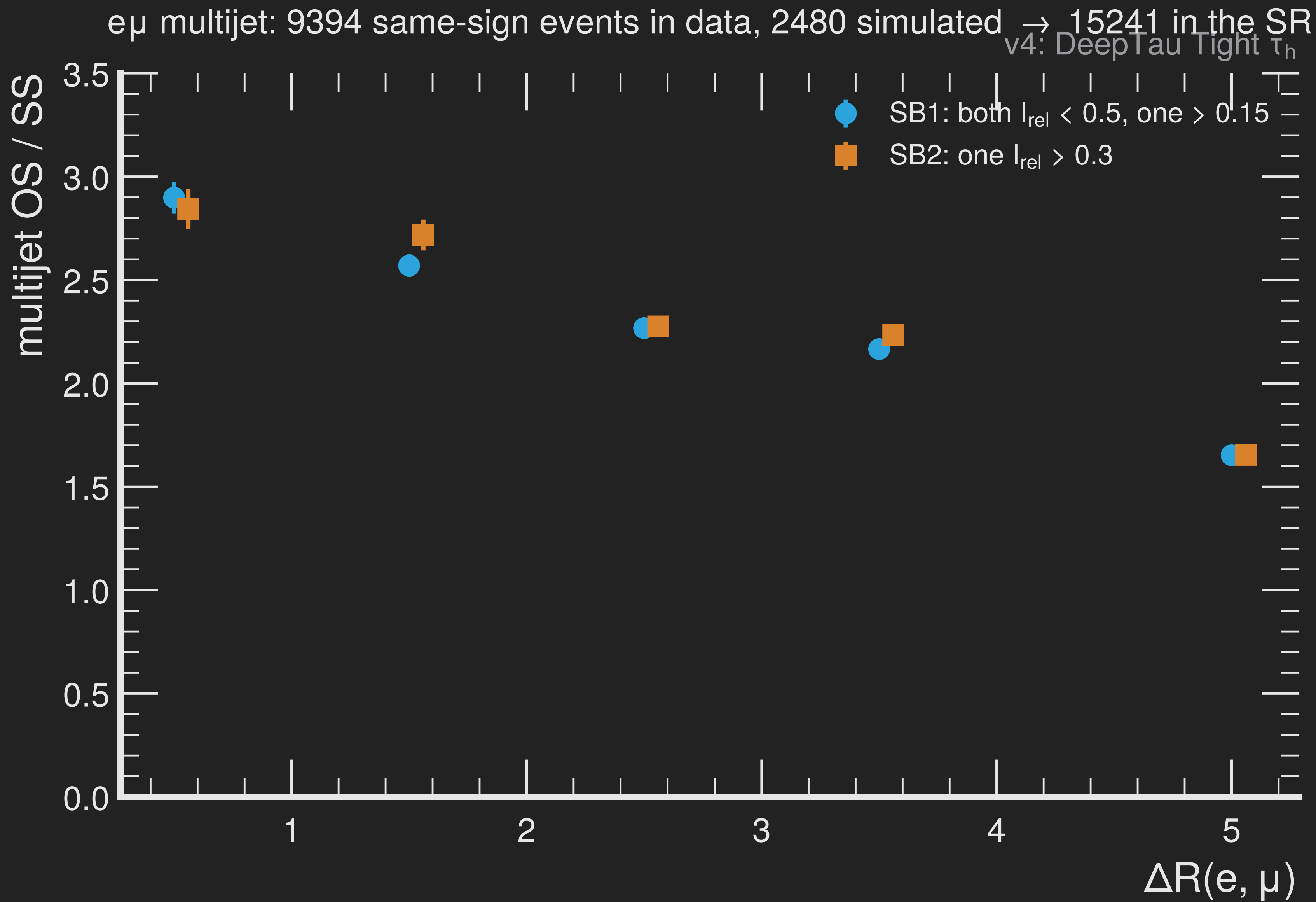
5. BACKGROUNDS MEASURED FROM DATA

$\mu\tau_h$: per-process fake factors — $C(\text{OS/SS}) = 1.05 \pm 0.03$, same-sign closure 0.966 ± 0.014

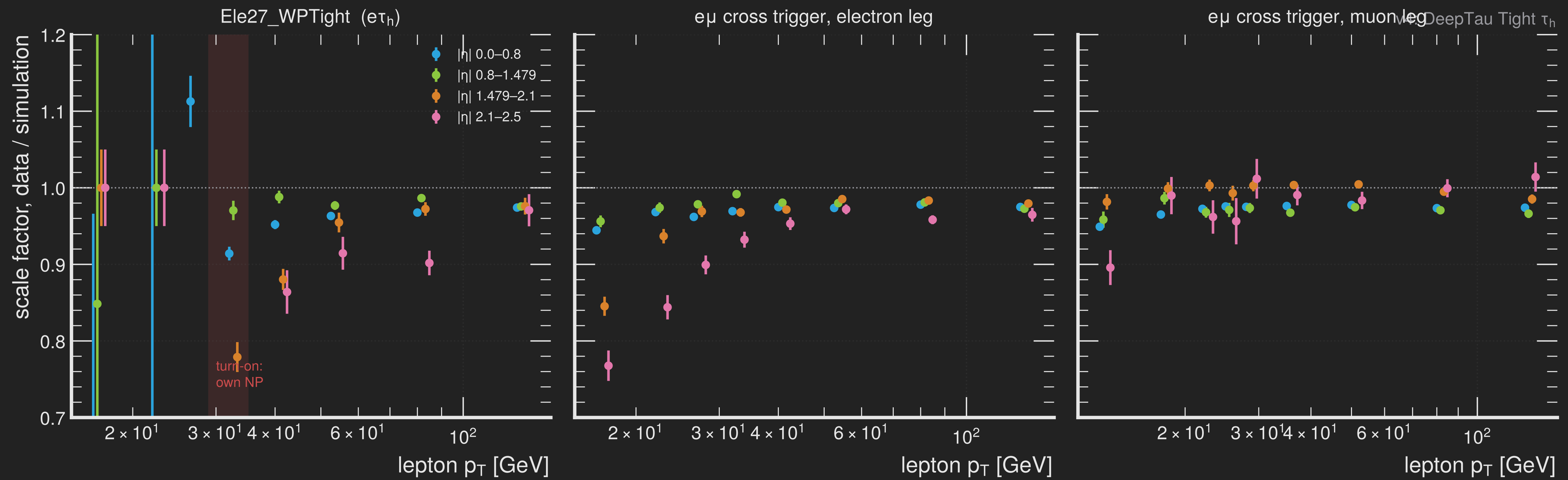


e\tau_h: per-process fake factors — C(OS/SS) = 1.14 ± 0.14, same-sign closure 1.025 ± 0.021





6. TRIGGER EFFICIENCIES, MEASURED IN SITU



7. FOR THE COMBINATION

deliverables, all committed

what	path	note
fit inputs, all channels	fit/fitinputs/ztautau.root	13 fitted regions, 62 systematics
fit inputs, per channel	fit/fitinputs/ztautau_<ch>.root	histograms copied bit for bit
configs	fit/ztautau[_<ch>].config	POI mu_Z, scale factors free
workspaces	fit/results/ztautau[_<ch>]/RooStats/	what a MultiFit reads
the numbers	output/results.json -> for_combination	sigma_pred, groups, ranking
the instructions	docs/11-combination-inputs.md	read section 6 before running

six things that will bite: docs/11-combination-inputs.md sections 5-7

	what it is	what to do
signal is 15 templates	DYtautau_tDM*, listed in the metadata as signal_samples	mu_Z goes on all of them, not on a sample called DYtautau
no acceptance parameters	acceptance_in_fit = true	the templates already vary A x eff inside the fit: Acc_* double counts
empty bins	tautau_SR2 bin 1 is dropped	its unconstrained gamma made the offset log(0)
e mu overlaps z-mumu	our emu_SR shares events with mumu_CRemu	one of the two must be dropped
muon / electron NPs	ours are the POG jsons	decorrelate unless both groups agree they are the same
ttbar	we float mu_ttbar from emu_CRtt	z-mumu constrains it with XS_TTbar: decide once